Optimization assignment

100 raw points · Individual submission · Rev 3

Purpose: use mathematical certificates, algorithm comparisons, modeling diagnosis and sensitivity analysis to support decisions. The final exam assesses foundations and short calculations; this assignment assesses the quality of your analysis and verification.

Due December 2, 2026. The course-grade weighting will be announced separately.

Submit one PDF report with numbered answers and one runnable notebook or Python script. Include mathematical work, software versions, solver status, objective values and independent feasibility checks. A screenshot of a solver answer alone is insufficient.

You may consult course materials and documentation. Explain your own reasoning, cite external assistance and disclose any AI or code assistance. Do not submit another student's derivations or results. Aim for 8–12 report pages including figures, plus runnable code and full trace files. This is a suggested length, not a scoring criterion. State assumptions and justify conclusions; solver output alone does not establish optimality.

1. LP certificate and sensitivity · 25 points

Maximize $4x+3y$ subject to $x+y\le5$, $2x+y\le8$ and $x,y\ge0$.

(a) (4 points) Form the dual using nonnegative resource prices u,v and explain the units of each price if the objective is period profit and each RHS is a resource quantity.

(b) (6 points) Find primal and dual optima at $b_1=5$. Verify both feasibility conditions, equality of objectives and every complementary-slackness product. Explain why this is a global certificate.

(c) (9 points) Vary only $b_1\ge0$. Derive the complete piecewise optimal solution and value function $V(b_1)$, with a dual certificate for every regime. Derive the closed interval where the original prices remain optimal. Plot V and identify its one-sided slopes at both breakpoints; distinguish an optimal endpoint multiplier from a unique two-sided derivative.

(d) (6 points) Predict and reoptimize at $b_1=5.5$ and 9. Quantify the error of each original-price prediction. Write a short resource-purchase recommendation explaining when price-based extrapolation is reliable.

2. Integer production and start-up cost · 20 points

Maximize $P=7a+10b-2z$ subject to $2a+3b\le10$, $a\le4z$, a,b nonnegative integers, and $z\in\{0,1\}$. The variables a,b count batches. When enabled, product a has a separate operating limit of4 batches. Only product a requires the start-up decision z, which incurs a fixed cost of2 over this planning period.

(a) (4 points) Explain both implications of $a\le4z$. Does $z=1$ force $a>0$? Explain why $M=4$ is valid for this model.

(b) (5 points) Form and solve the LP relaxation by allowing $a,b\ge0$ to be real and $0\le z\le1$. Give its optimum and a mathematical upper-bound certificate.

(c) (4 points) Before using exhaustive enumeration, verify the candidate $(a,b,z)=(2,2,1)$ and use its value with the LP bound to bracket the integer optimum. Can these bounds alone certify the candidate as optimal?

(d) (7 points) Complete bounded enumeration for both values of z using the largest feasible integer b for each a. Certify the global optimum. Then suppose a programmer replaces $a\le4z$ by $a\le5z$ and drops the separate four-batch operating limit. Solve this altered model and explain whether a higher reported profit represents an algorithmic improvement or a modeling error.

3. Unconstrained iterations and Armijo · 20 points

Minimize $f(u,v)=(u-1)^2+2(v+1)^2$ from $r_0=(3,1)$. Use Euclidean gradient norms. Armijo accepts equality: $f(r+\alpha p)\le f(r)+c_1\alpha\nabla f(r)^Tp$. In the nonconvex extension use $q(s)=(s^2-1)^2$ from $s_0=1/4$.

(a) (3 points) Derive the gradient and Hessian of f and certify its unique global minimizer.

(b) (3 points) At r_0, compute a negative-gradient step with $\alpha=0.1$. Independently derive the exact minimizer $\alpha\ge0$ of $f(r_0+\alpha p_0)$, where $p_0=-\nabla f(r_0)$. Report and compare both new points and objective values.

(c) (4 points) Restart at r_0 and apply Armijo with $c_1=0.1$, $\rho=0.5$ and initial $\alpha=1$. Use exact arithmetic to show each candidate point, actual value, target and decision through the first acceptance. Explain the equality case.

(d) (5 points) Implement fixed-step $\alpha=0.1$ and Armijo from the same r_0. Recompute the gradient and reset the initial Armijo α to1 each outer iteration. Stop before taking a step when $\|\nabla f\|_2<10^{-8}$; allow at most100 accepted steps and40 trials per search. For each iterate record point, gradient, norm and objective; for each update record accepted α and trial count. Submit a full trace and a plot of gradient norm against accepted steps. Count objective evaluations consistently: count $f(r_0)$ once, cache current values, and count every trial evaluation; exclude analytic exact-line-search work. Compare convergence and evaluation cost, and explain why two-step Armijo termination is special to this problem. Label any zero norm explicitly in a log-scale plot.

(e) (5 points) For q at s_0, compute q', q'', the raw Newton direction and its directional derivative. Determine whether sufficiently small positive steps in that direction can satisfy Armijo with $c_1=0.1$, and justify your conclusion. Propose a safeguarded descent direction, then run the same backtracking rule for one accepted update and verify the result. Explain what this example shows about using Newton on a nonconvex objective.

4. Constrained NLP and a capacity decision · 35 points

Minimize $F(x,y)=(x-4)^2+2(y-3)^2$ subject to $x+y\le b$, $x,y\ge 0$, and $b\ge 0$. Initially $b=4$. Use $g_1=x+y-b$, $g_2=-x$, $g_3=-y$ and $L=F+\Sigma\lambda_i g_i$. Treat F as a monetary operating loss over one planning period. Extra-capacity charges in part(d) apply to that same period.

(a) (8 points) Derive the candidate with x,y>0 and active capacity. Compute all multipliers and F at b=4.

(b) (6 points) Verify every KKT condition at b=4 and explain whether it certifies a unique global optimum.

(c) (7 points) Derive the complete parameter interval on which the capacity-active formula is feasible and satisfies KKT. Separately identify where x,y>0 and where $\lambda_1>0$. Explain what changes at each endpoint.

(d) (9 points) First evaluate the indivisible offer b:4→4.5 priced at3.8 per added unit for the same period. Compare exact operating-loss reduction, first-order prediction, cost and net benefit, then accept or reject it. Next suppose capacity can instead be purchased continuously: choose $0\le q\le 0.5$, set $b=4+q$, and pay3.8q. Derive and globally optimize exact net benefit, compare with q=0 and0.5, and interpret why the continuous and indivisible decisions differ. Plot exact net benefit and its linear approximation over this interval.

(e) (5 points) At b=4, run an NLP solver from (0,0), (1,1) and (4,3); the last start is intentionally infeasible. For each run report success/message, point and objective. Independently report $\max(0,\max_i g_i)$, $\max(0,\max_i(-\lambda_i))$, $\|\nabla F+\Sigma\lambda_i\nabla g_i\|_\infty$ and $\max_i|\lambda_i g_i|$. State how multipliers are obtained and the tolerances used. Diagnose any failed run.

Submission checklist

All four parts are answered. Figures label axes and units. Code runs from top to bottom. Results are compared with analytic calculations. Sensitivity conclusions state the parameter change and valid interval. Attach source code and any data you created.